

```
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find().pretty()
[
  {
    _id: ObjectId('6a83d93ef8d69a3c73adc982'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  }
]
```

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Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find({
|   avgRating: { $gt: 4 },
|   formats: { $in: ["Digital"] }
| })
[
  {
    _id: ObjectId('6a83d93ef8d69a3c73adc982'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drw: true,
      narrator: 'M. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      },
      {
        member_id: 'M1042',
        rating: 4,
        comment: 'Good but the Digital edition had formatting issues.',
        date: ISODate('2023-02-20T00:00:00.000Z')
      }
    ]
  }
]
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { formats: { $regex: "Digital", $options: "i" } },
|   { title: 1, formats: 1 }
| )
[
  {
    _id: ObjectId('6a83d93ef8d69a3c73adc982'),
    title: 'The Silent Algorithm',
    formats: [ 'Hardcover', 'Digital', 'Audiobook' ]
  }
]
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { "reviews.comment": { $regex: "Digital", $options: "i" } },
|   { title: 1, "reviews.$": 1 }
| )
[
  {
    _id: ObjectId('6a83d93ef8d69a3c73adc982'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: ISODate('2023-01-15T00:00:00.000Z')
      }
    ]
  }
]

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Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])
[
  {
    totalReviews: 2,
    author: 'R. Menon',
    title: 'The Silent Algorithm',
    avgReviewRating: 4.5
  }
]

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Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.createIndex({ "reviews.comment": "text" })
reviews.comment_text
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { $text: { $search: "Digital" } },
|   { title: 1, score: { $meta: "textScore" } }
| ).sort({ score: { $meta: "textScore" } })
|
|   {
|     _id: ObjectId('6a83d93ef8d69a3c73adc982'),
|     title: 'The Silent Algorithm',
|     score: 1.1833333333333333
|   }
| ]
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
|   { $text: { $search: "Digital" } }
| ).explain("executionStats")
|
| {
|   explainVersion: '1',
|   queryPlanner: {
|     namespace: 'BookFlow.book_metadata',
|     parsedQuery: {
|       '$text': {
|         '$search': 'Digital',
|         '$language': 'english',
|         '$caseSensitive': false,
|         '$diacriticSensitive': false
|       }
|     },
|     indexFilterSet: false,
|     queryHash: 'CF6F4CEE',
|     planCacheShapeHash: 'CF6F4CEE',
|     planCacheKey: '08852285',
|     optimizationTimeMillis: 0,
|     maxIndexedOrSolutionsReached: false,
|     maxIndexedAndSolutionsReached: false,
|     maxScansToExplodeReached: false,
|     prunedSimilarIndexes: false,
|     winningPlan: {
|       isCached: false,
|       stage: 'TEXT_MATCH',
|       indexPrefix: {},
|       indexName: 'reviews.comment_text',
|       parsedTextQuery: {
|         terms: [ 'digit' ],
|         negatedTerms: [],
|         phrases: [],
|         negatedPhrases: []
|       },
|       textIndexVersion: 3,

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    inputStage: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { _fts: 'text', _ftsx: 1 },
        indexName: 'reviews.comment_text',
        isMultiKey: true,
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {}
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 1,
    executionTimeMillis: 0,
    totalKeysExamined: 1,
    totalDocsExamined: 1,
    executionStages: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      indexPrefix: {},
      indexName: 'reviews.comment_text',
      parsedTextQuery: {
        terms: [ 'digit' ],
        negatedTerms: [],
        phrases: [],
        negatedPhrases: []
      },
      textIndexVersion: 3,
      docsRejected: 0,
      inputStage: {
        stage: 'FETCH',
        nReturned: 1,

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        executionTimeMillisEstimate: 0,
        works: 2,
        advanced: 1,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        docsExamined: 1,
        alreadyHasObj: 0,
        inputStage: {
          stage: 'IXSCAN',
          nReturned: 1,
          executionTimeMillisEstimate: 0,
          works: 2,
          advanced: 1,
          needTime: 0,
          needYield: 0,
          saveState: 0,
          restoreState: 0,
          isEOF: 1,
          keyPattern: { _fts: 'text', _ftsx: 1 },
          indexName: 'reviews.comment_text',
          isMultiKey: true,
          isUnique: false,
          isSparse: false,
          isPartial: false,
          indexVersion: 2,
          direction: 'backward',
          indexBounds: {},
          keysExamined: 1,
          seeks: 1,
          dupsTested: 1,
          dupsDropped: 0
        }
      }
    }
  },
  queryShapeHash: 'B706EBB3DB671ACEA1F4599BFB25D4E8CE972167EC9201BE774DFF1935DAA2FA',
  command: {
    find: 'book_metadata',
    filter: { '$text': { '$search': 'Digital' } },
    '$db': 'BookFlow'
  },
  serverInfo: {
    host: 'ac-bo4lwn1-shard-00-01.voy9vg9.mongodb.net',
    port: 27017,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
  }
}

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},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787026069, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('JQJw5m5rfqU4jadu15jKJMkETNQ=', 0),
    keyId: Long('7617890356549386242')
  }
},
operationTime: Timestamp({ t: 1787026069, i: 1 })
}
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
| { "reviews.comment": { $regex: "Digital", $options: "i" } }
| ).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    indexFilterSet: false,
    queryHash: '9DF7CB16',
    planCacheShapeHash: '9DF7CB16',
    planCacheKey: 'A9183C86',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
      direction: 'forward'
    },
    rejectedPlans: []
  },
  executionStats: {

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```
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 0,
  totalKeysExamined: 0,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    direction: 'forward',
    docsExamined: 1
  }
},
queryShapeHash: 'E85AE31DF35193E59E333508E2FF982C9EF836C91317C9F6A34A0EEB6597894A',
command: {
  find: 'book_metadata',
  filter: { 'reviews.comment': { '$regex': 'Digital', '$options': 'i' } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'ac-bo4lwn1-shard-00-01.voy9vg9.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
```

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OK: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787026075, i: 3 }),
  signature: {
    hash: Binary.createFromBase64('8odOWOvMnobqGWuykLezQ405soA=', 0),
    keyId: Long('7617890356549386242')
  }
},
operationTime: Timestamp({ t: 1787026075, i: 3 })

Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.createIndex({ formats: 1 })
formats_1
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.createIndex({ avgRating: -1 })
avgRating_-1
Atlas atlas-4aubha-shard-0 [primary] BookFlow> db.book_metadata.find(
  { avgRating: { $gt: 4 } }
).explain("executionStats")

explainVersion: '1',
queryPlanner: {
  namespace: 'BookFlow.book_metadata',
  parsedQuery: { avgRating: { '$gt': 4 } },
  indexFilterSet: false,
  queryHash: '85BA8276',
  planCacheShapeHash: '85BA8276',
  planCacheKey: '6EB4A3FD',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'FETCH',
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultiKey: false,
      multiKeyPaths: { avgRating: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { avgRating: [ '[inf.0, 4)' ] }
    }
  },
  rejectedPlans: []
}

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},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 1,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 1,
    executionTimeMillisEstimate: 1,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    docsExamined: 1,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 1,
      executionTimeMillisEstimate: 1,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      keyPattern: { avgRating: -1 },
      indexName: 'avgRating_-1',
      isMultiKey: false,
      multiKeyPaths: { avgRating: [] },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: { avgRating: [ '[inf.0, 4)' ] },
      keysExamined: 1,
      seeks: 1,
      dupsTested: 0,
      dupsDropped: 0
    }
  }
}
},

```

```
queryShapeHash: '992969EA42A38D110981577D390322577B56F065886CE3C4D7D88FB1C4779AE9',
command: {
  find: 'book_metadata',
  filter: { avgRating: { '$gt': 4 } },
  '$db': 'BookFlow'
},
serverInfo: {
  host: 'ac-bo4lwn1-shard-00-01.voy9vg9.mongodb.net',
  port: 27017,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 16793600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 33554432,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1787026090, i: 9 }),
  signature: {
    hash: Binary.createFromBase64('Ij/7lXgjlqN0y6loLXJsYMMqGw0=', 0),
    keyId: Long('7617890356549386242')
  }
},
operationTime: Timestamp({ t: 1787026090, i: 9 })
}
Atlas atlas-4aubha-shard-0 [primary] BookFlow> |
```